GRISHMA BALGI

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Computer Vision & Robot Perception

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Programming: Python, NumPy, Pandas, C++

Vision: RootSIFT, VLAD, visual place recognition, MediaPipe

Platforms: Raspberry Pi, Jetson Nano, MuJoCo; PyTorch/TensorFlow exposure

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.
- Developing an 18-segment model of a fabricated TPU 95A tentacle in MuJoCo, representing passive bending with effective joint stiffness, damping and tendon elements.
- Iterating on geometry and passive response; investigating fluid-force effects while separating provisional model parameters from quantities requiring physical calibration.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Worked on reduced-order modelling of mechanical systems using Creo and Altair tools, preparing simulation-derived data and exploring models for circuit-breaker mechanisms.
- Built experience in sensor selection, vibration data acquisition and Python-based analysis for circuit-breaker condition monitoring.

SELECTED PROJECTS

Camera-Only Visual Navigation
Python, RootSIFT, VLAD, graph search | Course project | In progress

- Building a camera-only navigation pipeline using visual descriptors, a topological graph and Dijkstra search to navigate toward image-defined goals.
- Working on place recognition and geometric verification; evaluating extensions separately from the baseline rather than treating proposed methods as completed results.

Baymax | Desk Companion Robot
Python, Raspberry Pi, Jetson Nano | Course project

- Developed a desk-companion robot using Raspberry Pi and Jetson Nano, combining camera and voice inputs with motor, LED and display outputs.
- Structured the application around input processing, conversational responses and physical feedback; implemented hardware and software components within a team project.